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Digital Forensics and Cyber Law

Antara Arifa Mullick
210042132
Program: B.Sc. in SWE
Department of CSE

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Contents

1 The “Retail Ransom”: Shwapno Database Breach **2**

2 The “SpiderKash” Leak: FBI Director’s Gmail Breach **6**

3 Digital Forensics Tool: FTK Imager **10**

Digital Forensics and Cyber Law

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1 The "Retail Ransom": Shwapno Database Breach

Scenario: Shwapno, a major Bangladeshi retail chain, suffers unauthorised access to a database of 4 million customer records. Attackers exfiltrate the data and demand a $1.5 million ransom to prevent publication on the dark web.

Q1 — How can forensics determine whether data was exfiltrated or only encrypted in place?

Forensic investigators distinguish exfiltration from in-place encryption by examining several complementary evidence sources:

- **Network flow logs and firewall records:** The most direct indicator. Investigators review NetFlow data, proxy logs, and NGFW session logs for large outbound transfers — particularly to unusual external IP addresses or geographies — in the hours or days before the ransom note appeared. A spike in egress traffic that cannot be attributed to normal business operations is strong evidence of exfiltration [1].
- **DLP and SIEM alerts:** If a Data Loss Prevention solution was deployed, alert logs may record policy violations corresponding to bulk database exports. SIEM correlation rules may also flag anomalous `SELECT *` queries or bulk `mysqldump`/`pg_dump` executions under unusual accounts.
- **Database audit logs:** Database-level audit trails record every query executed. Mass read operations on customer tables by accounts not normally performing reporting functions indicate data was accessed in bulk before or alongside encryption.
- **File system staging artefacts:** Attackers frequently stage exfiltrated data in temporary archive files (`.zip`, `.rar`, `.7z`) before transmission. Forensic imaging may reveal residual MFT records corresponding to large archive creation even after deletion.
- **Dark web and threat intelligence:** If a sample of the stolen data appears in dark-web posts or ransomware leak sites as proof-of-exfiltration, this constitutes external corroboration that data left the network.

Q2 — First Responder Actions

As the first responder, I would follow the standard IR lifecycle while prioritising evidence preservation over remediation speed:

1. **Isolate without powering off:** Immediately segment the compromised server from the network using VLAN isolation or ACL blocks to prevent ongoing exfiltration. I would *not* power down the system, as volatile memory may contain running malware, encryption keys, or attacker session tokens that are lost on shutdown.

2. **Capture volatile evidence first:** Using a write-blocker and forensic tools (e.g., Magnet RAM Capture), acquire a full memory image. Then capture running processes, active network connections (`netstat -anb`), and logged-in users before any changes are made [2].

3. **Preserve and hash all logs:** Collect and cryptographically hash (SHA-256) all relevant logs — firewall, proxy, database audit, and authentication logs — before they are overwritten by rotation policies.

4. **Forensic disk imaging:** Create a bit-for-bit forensic image using FTK Imager. Verify integrity with MD5/SHA-256 hashes. Work exclusively on the image copy; never on the original evidence.

5. **Identify the initial access vector:** Review authentication logs for brute-force attempts, credential stuffing, or anomalous logins. Check for recently installed web shells, modified scheduled tasks, or new service entries.

6. **Notify stakeholders and legal counsel:** Alert senior management and the legal team and — if required by Bangladesh's Cyber Security Act 2023 — the relevant regulatory authority. Document all actions with timestamps.

7. **Do not pay the ransom immediately:** Advise management to defer any payment decision pending legal review and an assessment of backup recovery options.

Q3 — Recovery Activities Before Considering the Ransom

Before any ransom payment is considered, the following technical recovery approaches should be exhausted in order of least to most disruptive:

1. **Check offline and cloud backups:** Verify whether clean, pre-attack backups exist. Air-gapped backups that ransomware could not reach may allow immediate restoration. Cloud provider snapshots (e.g., AWS RDS, Azure SQL) should also be checked.

2. **Volume Shadow Copy (VSS) recovery:** On Windows systems, VSS maintains point-in-time volume snapshots. If attackers did not execute `vssadmin delete shadows`, previous versions of the database can be mounted and restored using ShadowExplorer [3].

3. **Ransomware decryptor lookup:** Check the **No More Ransom** project (`https://www.nomoreransom.org`), a Europol/INTERPOL collaboration providing free decryptors for many known ransomware families [4].

4. **File carving from unallocated space:** Using tools such as **Autopsy** or **PhotoRec**, attempt to recover deleted or partially overwritten database files from unallocated disk space. Ransomware typically encrypts the active file and marks the original deleted; carving may recover pre-encryption versions if sectors have not been overwritten.

5. **Database transaction log replay:** If transaction logs (`.ldf` in SQL Server, `binlog` in MySQL) survived on a separate volume, replay them against an older backup to reconstruct the database to a point close to the attack.

6. **Memory forensics for encryption keys:** In some ransomware variants, the symmetric encryption key is briefly held in memory. Analysis of the acquired RAM image using **Volatility** may allow key extraction and decryption without payment.

Q4 — Does Paying the Ransom Constitute Aiding and Abetting? Legal Risks of Payment

The Aiding and Abetting Question

Whether paying a ransom constitutes criminal "aiding and abetting" depends on jurisdiction, but the legal risks are substantial and growing:

- **Bangladesh ICT Act 2006 (as amended) and Cyber Security Act 2023:** Under Section 54 of the ICT Act 2006, any person who "abets" the commission of a computer-related offence is liable to the same punishment as the principal offender. A ransom payment that directly finances further criminal activity by the same group could, in principle, be characterised as abetment under this provision [7]. More practically, under the Cyber Security Act 2023, Section 17 imposes mandatory reporting obligations on organisations that suffer notifiable cyber incidents to the Bangladesh Computer Council (BCC). Paying a ransom while concealing the breach from the BCC would constitute a separate offence of failing to report a notifiable incident, carrying fines and potential imprisonment [6].
- **U.S. OFAC sanctions exposure (relevant for international context):** The U.S. Treasury's OFAC has explicitly warned that ransomware payments to sanctioned entities may violate the International Emergency Economic Powers Act (IEEPA), regardless of whether the payer knew the recipient was sanctioned. A 2020 OFAC advisory stated that facilitating ransomware payments may also expose intermediaries to sanctions liability [5]. This is relevant because many Bangladeshi companies have U.S. dollar-denominated banking relationships; making a cryptocurrency ransom payment to a sanctioned group could implicate U.S. secondary sanctions.
- **Financing of terrorism / money laundering:** If the ransomware group has known links to state-sponsored or terrorist organisations, payment could also engage Bangladesh's Anti-Terrorism Act 2009 and Money Laundering Prevention Act 2012, both of which prohibit the provision of funds to designated entities.

Long-Term Legal Risks of Payment

1. **No guarantee of non-publication:** Paying does not legally or practically bind the attacker to delete the data. In double-extortion schemes, groups have published or resold data even after receiving payment. The company still faces full regulatory liability for the breach regardless of payment.
2. **Regulatory and civil liability:** Under data protection frameworks (Bangladesh's proposed Personal Data Protection Act, and GDPR for any customers whose data falls within its scope), the breach itself triggers notification and remediation obligations. Ransom payment does not discharge these.
3. **Increased re-targeting:** Organisations that pay ransoms are re-targeted at significantly higher rates, as payment signals willingness and financial capacity, creating compounding long-term risk.
4. **Director liability:** In some jurisdictions, directors who authorise payments to sanctioned entities without adequate due diligence face personal civil or criminal liability separate from corporate liability.

Q5 — Legal Liability if the Database Was Unencrypted

If forensic investigation reveals that Shwapno stored personal data of 4 million customers in an unencrypted database, the company faces substantial legal liability *independent of and in addition to* any liability attributable to the hackers.

The "Adequate Security Measures" Standard

- **GDPR (Art. 32):** Requires "appropriate technical and organisational measures" including, where appropriate, "encryption of personal data." Failure to encrypt a large-scale customer database could attract fines of up to 4% of annual global turnover or €20 million, whichever is higher. GDPR may apply extraterritorially if any EU residents' data is processed [9].
- **Bangladesh Cyber Security Act 2023 / proposed Personal Data Protection Act:** These frameworks place an obligation on data processors to maintain security of personal data. Storing sensitive customer records in plaintext constitutes a material failure of this obligation [6].
- **ISO/IEC 27001:** Courts and regulators frequently reference international standards when assessing reasonableness. Unencrypted storage of this scale would not satisfy ISO 27001 Annex A.8.24 (use of cryptography) requirements.

The Hacker's Actions Do Not Extinguish the Company's Liability

A critical legal principle applies here: the intervening criminal act of a third party does not automatically break the chain of causation between the company's negligent security practices and the harm suffered by data subjects. Courts in data protection cases consistently hold that where a breach was *foreseeable* and the company's failure to implement basic security measures *materially facilitated* the harm, the company remains liable.

In plain terms: the hacker is criminally responsible for the intrusion; the company is legally responsible for making the impact catastrophically worse by failing to encrypt the data it was entrusted to protect.

Liability to Customers: Negligence and Consumer Rights

Beyond regulatory fines, Shwapno faces direct civil liability to the 4 million affected customers under two theories:

- **Negligence (tort):** Customers whose personal data was exposed can bring negligence claims arguing that Shwapno owed them a duty of care in handling their data, breached that duty by storing it unencrypted, and caused foreseeable harm (identity theft risk, financial fraud, emotional distress). Bangladesh's Consumer Rights Protection Act 2009 (CRPA) further provides that consumers harmed by a seller's failure to maintain adequate service standards may seek compensation through the Directorate of National Consumer Rights Protection (DNCRP) or civil courts [8].
- **Aggravated regulatory penalties for scale:** The combination of unencrypted storage, failure to report, and the scale of 4 million affected individuals would likely attract the maximum tier of penalties available under both the Cyber Security Act 2023 and any future

Personal Data Protection Act, in addition to reputational damage that would materially affect Shwapno's commercial relationships.

2 The "SpiderKash" Leak: FBI Director's Gmail Breach

Scenario: On March 27, 2026, pro-Iranian group Handala claimed to have breached the personal Gmail account of FBI Director Kash Patel, leaking personal photos, his CV, and over 300 emails (2010–2019) in a dossier titled "SpiderKash."

Q1 — Using DKIM Signatures to Authenticate Leaked Emails

DomainKeys Identified Mail (DKIM) is a cryptographic email authentication standard (RFC 6376) that allows a forensic investigator to verify that a specific email message was authorised by the sending domain and that its content has not been altered in transit.

How DKIM Works

When Google's mail servers send a message, they compute a cryptographic hash of specified header fields and the message body, then sign it with a **private RSA or Ed25519 key** held by Google. The resulting digital signature is embedded in the `DKIM-Signature` header. Google publishes the corresponding **public key** in its DNS records at a selector subdomain (e.g., `20230601._domainkey.gmail.com`).

Forensic Verification Procedure

1. **Extract the raw email headers:** Retrieve the complete `.eml` file, preserving the full `DKIM-Signature` header including the signing domain (`d=gmail.com`), selector (`s=`), and base64 signature (`b=`).

2. **Retrieve Google's public key from DNS:** Query the DNS TXT record at the selector address using `dig` or `nslookup`. This key is controlled by Google and cannot be fabricated by Handala.

3. **Recompute and verify the signature:** Using a tool such as **dkimverify** (Python), apply the same canonicalisation algorithm to the signed headers and body, compute the hash, and verify it against the signature using Google's public key. A valid signature proves: (a) the email was sent through Google's infrastructure; and (b) the content has not been modified since signing [10].

4. **Limitation:** DKIM authenticates the sending infrastructure and content integrity but does *not* prove which human composed the message. Additional corroboration from Google's account activity records (via legal process) would be needed to attribute the specific Gmail account.

If the leaked emails were fabricated and typed in a Word document, they would lack a valid DKIM signature verifiable against Google's live DNS public key, immediately exposing them as forgeries.

Q2 — Cloud Acquisition vs. Physical Acquisition

Traditional Physical Acquisition (Laptop Hard Drive)

1. Attach a **hardware write-blocker** to prevent writes during acquisition.
2. Create a **bit-for-bit forensic image** using FTK Imager or `dd`, capturing every sector including deleted files and unallocated space.
3. Generate and record **MD5 and SHA-256 hash values** to verify integrity.
4. Work exclusively on a verified copy; preserve the original as evidence.

This produces a complete, verifiable copy of all data including deleted files, slack space, and metadata, and is well-established in courts as forensically sound [11].

Cloud Acquisition (Gmail Account)

1. **Legal process (court order / MLAT):** Obtain a court order or subpoena compelling Google to preserve and produce account data under the Stored Communications Act (18 U.S.C. § 2703). In cross-border cases, a Mutual Legal Assistance Treaty (MLAT) request may be required.
2. **Preservation request:** Under 18 U.S.C. § 2703(f), law enforcement can send an emergency preservation request requiring Google to retain account data for 90 days while legal process is obtained.
3. **API-based forensic acquisition:** Tools such as **AXIOM Cyber** (Magnet Forensics) or **Oxygen Forensic Detective** use OAuth-authenticated Gmail API calls to systematically export messages, metadata, and access logs. Each item is hashed upon acquisition to establish integrity.
4. **Account activity and access logs:** Google retains IP access logs, device sign-in records, and OAuth token histories. These establish *when* and *from where* the account was accessed by Handala, independent of email content [12].

The key limitation of cloud acquisition is complete dependence on the provider's cooperation and data retention policies; the investigator cannot independently verify that all data has been produced.

Q3 — Using EXIF Metadata to Verify Timeline and Provenance of Leaked Photos

EXIF (Exchangeable Image File Format) metadata is embedded within image files and records technical and contextual data about each photograph. Key forensic fields:

- `DateTimeOriginal`: The date and time the shutter was pressed.
- `GPSLatitude` / `GPSLongitude`: GPS coordinates at capture.
- `Make` / `Model`: Camera or smartphone manufacturer and model.
- `Software`: Any software that processed the image (e.g., Lightroom).

- `FileModifyDate`: Last file system modification — distinguishes cloud-synced from physical backup copies.

Forensic Application to SpiderKash

1. **Verify internal consistency:** Extract EXIF data using **ExifTool** or Autopsy's metadata module. If `DateTimeOriginal` reads 2015 and GPS coordinates place the device at the claimed venue, this corroborates provenance.
2. **Cross-reference GPS coordinates:** Plot GPS tags on a map to verify whether the location matches the claimed event. Public event records and news photographs from the same date provide independent corroboration.
3. **Distinguish cloud sync from physical backup:**
   - A **cloud-synced photo** (Google Photos) typically has `DateTimeOriginal` matching the 2015 capture time, but `FileModifyDate` matching the cloud upload time.
   - A **physical backup copy** (USB drive) shows `FileModifyDate` matching the time of the copy operation, significantly later than capture.
   - If Handala exfiltrated photos from Gmail's integrated Google Photos storage, the file system timestamps would reflect the March 2026 acquisition date while `DateTimeOriginal` remains 2015.
4. **Detect EXIF tampering:** Check for anomalies — `Software` tags referencing a metadata editor, or timestamps predating the device model's release date — which would indicate fabrication.

Q4 — Forensic Ethics and the Boundaries of a Digital Search

The Core Ethical Tension

When an investigator is authorised to search a personal account for "evidence of a hack," they inevitably encounter a vast body of private material — family photographs, medical information, personal correspondence — unrelated to the investigation. The legal and ethical framework governing what may be viewed, retained, and used is critical to both investigative integrity and the protection of fundamental rights.

Legal Boundaries: The Fourth Amendment Framework (U.S.)

The **Fourth Amendment** protects against "unreasonable searches and seizures" and requires warrants to "particularly describ[e] the place to be searched, and the persons or things to be seized." In *Riley v. California* (2014), the U.S. Supreme Court held unanimously that the government generally must obtain a warrant before searching the digital contents of a device, recognising that digital data contains "the privacies of life" [13].

A properly drafted warrant for the SpiderKash investigation would:

- Define a **specific scope** (e.g., "emails and attachments relating to unauthorised access, including login records, IP addresses, and communications with threat actors").
- Impose a **"plain view" limitation**: only evidence described in the warrant, or evidence of other crimes encountered in plain view during a lawful search, may be retained.

- Require **filter/taint team protocols**: a separate team reviews account contents first to segregate privileged or irrelevant personal materials before the investigation team sees them.

The "Plain View" Doctrine and Its Limits

An investigator authorised to search for evidence of a hack does *not* have blanket authorisation to read all personal photos and family emails. Under the plain view doctrine, incidentally encountered clear evidence of another crime may be seized. However, a forensic examiner who deliberately reads personal emails outside the warrant's scope has overstepped legal authorisation — any evidence so obtained may be suppressed, and the examiner may face personal disciplinary or legal consequences [14].

Equivalent Frameworks in Bangladesh and Beyond

Bangladesh's **Cyber Security Act 2023** (Sections 14–17) requires court authorisation for interception and monitoring of electronic communications, and imposes proportionality requirements on any digital search. Under the **Evidence Act 1872** (as still applied in Bangladesh), Section 93 requires that a search be conducted under a proper warrant issued by a Magistrate, specifying the place and the articles to be searched for. A digital search of a public official's personal account without proper judicial authorisation would render any evidence obtained inadmissible in Bangladeshi courts and could expose the investigator to criminal liability under Section 26 of the CSA 2023 (unauthorised access to a computer system) [6].

The EU's **GDPR** and **Law Enforcement Directive (LED)** impose additional constraints when personal data of EU residents is involved, requiring that any government access to personal data be *necessary*, *proportionate*, and subject to independent oversight.

Q5 — "Fruit of the Poisonous Tree": Admissibility of Illegally Obtained Leaked Evidence

The Fruit of the Poisonous Tree Doctrine

The **fruit of the poisonous tree** doctrine (derived from *Silverthorne Lumber Co. v. United States*, 1920, and *Wong Sun v. United States*, 1963) holds that evidence obtained as a direct result of an illegal *government* search is inadmissible in court. The rationale is deterrence: if illegally obtained evidence cannot be used, government actors have no incentive to violate constitutional protections [15].

Does It Apply to Third-Party Private Hacking?

Under U.S. constitutional law the answer is generally **no**. The Fourth Amendment applies to *government* conduct only. Where evidence is illegally obtained by a *private party* acting independently — not as a government agent — and the government subsequently receives that evidence, the exclusionary rule generally does not apply.

This was established in *United States v. Jacobsen* (1984), in which the Supreme Court held that the Fourth Amendment does not apply to a private person acting independently of government direction. If a private party or foreign hacking group independently obtains and publishes evidence, the government's use of it is not constitutionally barred — provided the government did *not direct, encourage, or ratify* the illegal search [16].

Application to the SpiderKash Scenario

If Handala independently breached Kash Patel's Gmail without any U.S. government direction or prior knowledge:

- The Fourth Amendment **does not** prohibit the government from receiving and using the leaked evidence in court.
- A federal prosecutor could use emails from the dossier as evidence of a financial crime, subject to authentication requirements (DKIM verification, Google records).
- However, the government **cannot independently access** the account beyond what Handala published without a separate warrant.

Important Caveats

1. **Authentication:** The prosecution must authenticate the leaked emails as genuine under Federal Rule of Evidence 901. DKIM verification and Google's preserved records are essential.
2. **Chain of custody:** Evidence that passed through a hacker's hands is inherently suspect for tampering; courts may require additional expert testimony to establish reliability.
3. **Government nexus:** If evidence emerges that U.S. government agents had advance knowledge of or any involvement in Handala's operation, the private-party exception would collapse and the evidence could be suppressed.
4. **The Inevitable Discovery Doctrine:** Even if a court were to find some government nexus, the prosecution could invoke the **inevitable discovery** exception established in *Nix v. Williams* (1984). Under this doctrine, unlawfully obtained evidence is nevertheless admissible if the government can demonstrate by a preponderance of the evidence that it would have *inevitably* discovered the same evidence through lawful means. If investigators could show they would have obtained the emails via a Google court order anyway, suppression may still be denied even if the Handala connection creates a legal problem [15].
5. **State law variations:** Some U.S. states have broader exclusionary rules under their own constitutions that may apply to privately obtained evidence in state prosecutions, regardless of the federal rule.

3 Digital Forensics Tool: FTK Imager

Tool: FTK Imager (Exterro/AccessData) — Industry-standard forensic acquisition and imaging utility.

Phase of the Forensic Process

FTK Imager is used in the **Evidence Acquisition and Preservation** phase of the digital forensics lifecycle. Under the **NIST SP 800-86** four-phase model, the forensic process consists of: (1) Collection, (2) Examination, (3) Analysis, and (4) Reporting. FTK Imager operates in **Phase 1: Collection** — the most legally critical phase, upon which all subsequent examination and analysis depends. Any compromise of integrity at this stage invalidates the entire downstream forensic process.

The **ACPO Good Practice Guide for Digital Evidence** similarly identifies four principles: Principle 1 requires that no action taken by law enforcement shall change data held on a digital device that may subsequently be relied upon in court. FTK Imager is specifically designed to satisfy this principle through write-blocked, hash-verified acquisition.

Core Functions

- **Forensic imaging:** Creates exact, bit-for-bit copies of storage media in industry-standard formats: **E01** (Expert Witness Format, with embedded metadata), **raw/dd** (uncompressed sector-by-sector copy), and **AD1** (logical image format).
- **Hash verification:** Computes **MD5 and SHA-1 hash values** of both the source media and resulting image. Matching hashes mathematically demonstrate that every bit was captured accurately and no modification occurred during imaging.
- **Write-blocker integration:** Operates with hardware write-blockers (e.g., Tableau, WiebeTech) that prevent any writes to the source drive, ensuring the original evidence is not contaminated.
- **Read-only live preview:** Mounts and browses drive contents without altering file system metadata (e.g., `atime` timestamps) that would otherwise be updated by normal OS file access.
- **RAM acquisition:** Captures volatile memory contents, preserving running processes, network connections, and encryption keys that would be lost on shutdown.

Ensuring Legal Admissibility

1. **Chain of custody documentation:** Generates a detailed case log recording the examiner's name, date and time of acquisition, device identifiers, and hash values — forming part of the chain of custody record required in court.
2. **Hash-based integrity verification:** The MD5/SHA-1 hash of the forensic image, independently verifiable by any party, proves the image presented in court is an exact copy of what was acquired. Any post-acquisition tampering produces a different hash, immediately exposing the alteration.
3. **Non-destructive operation:** By operating with write-blockers and never writing to the source media, FTK Imager satisfies the forensic principle that original evidence must be preserved unaltered. Courts have consistently held that write-blocked, hash-verified forensic images are admissible as reliable copies of the original [11].

FTK Imager's methodology aligns with **NIST SP 800-86** and the **ACPO Good Practice Guide for Digital Evidence**, making it one of the most widely court-accepted acquisition tools in both law enforcement and corporate forensic practice [1].

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